

AIMO Proof Pilot — Final Grade

Problem (#_ID): 2_ALTORO

Submission: model_4

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

The solution follows a coordinate approach but makes decisive errors early: the point $X = EF \cap BC$ is computed incorrectly, so the constructed line and hence Q are wrong. The endgame then silently assumes $PE = HE$ — there is no such relation, since P is the midpoint of HQ — which produces the wrong ratio ($11/8$, $a + b = 19$, against the correct $125/91$, $a + b = 216$). With the construction already broken and no partial-progress milestone reached, it scores **0**.